

Generative Proposals (Candidate Pool N_0)

Stage 1: Syntactic & AST Parsing

Checks: Syntax valid, elaborated module hierarchy, port types
Example engines: Tree-Sitter / Verilator / Slang • Relative cost: lowest

Stage 2: Interface Schema & Interconnect

Checks: AXI / TLM handshakes, port widths, clock domain tags
Example engines: Static Schema Linters / Protocol Checkers • Relative cost: low

Stage 3: Functional & Assertion Verification

Checks: Temporal invariants, state-machine deadlocks, coverage
Example engines: SVA checkers / BMC / simulators • Relative cost: moderate to high

Stage 4: Static Timing Analysis (STA)

Checks: Multi-corner setup & hold slack, WNS/TNS
Example engines: OpenSTA / PrimeTime / Tempus • Relative cost: high

Stage 5: Physical DRC / LVS Signoff

Checks: Design rules (DRC), layout vs. schematic (LVS)
Example engines: Calibre / KLayout / Netgen • Relative cost: highest

Signoff-Clean Implementation (GDSII / OASIS)

Near-Miss Timing & DRC Failures
Route to Localized ECO Optimization
(Buffer insertion, gate sizing, wire widening)